

Data

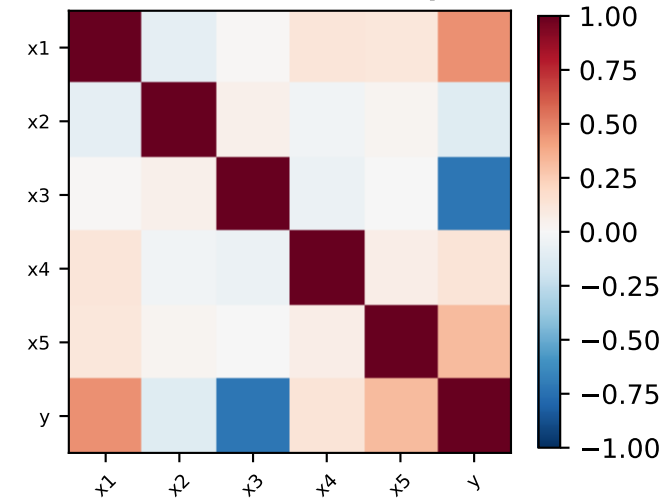
Sample Size & Structure

Training observations: 200
Test observations: 100
Predictors: x1, x2, x3, x4, x5
Response: y (continuous)

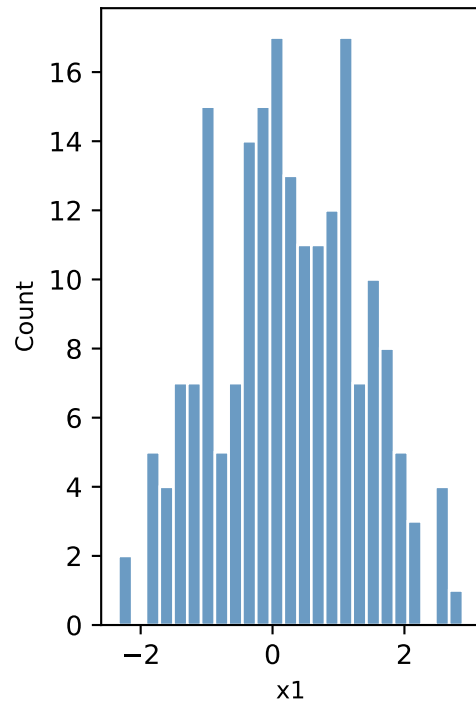
Summary Statistics

	x1	x2	x3	x4	x5	y
mean	0.24	-0.61	0.55	0.46	-0.12	0.95
std	1.09	0.94	1.23	1.07	0.79	2.18
min	-2.34	-3.43	-2.31	-2.44	-2.44	-5.34
max	2.89	1.93	3.55	3.71	1.96	7.39

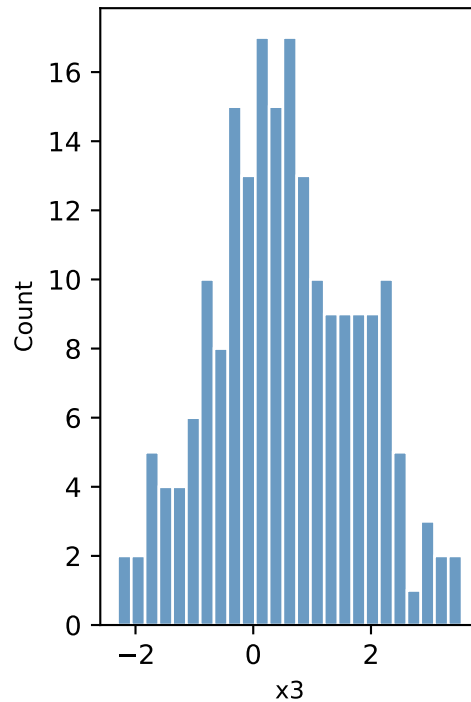
Correlation Heatmap



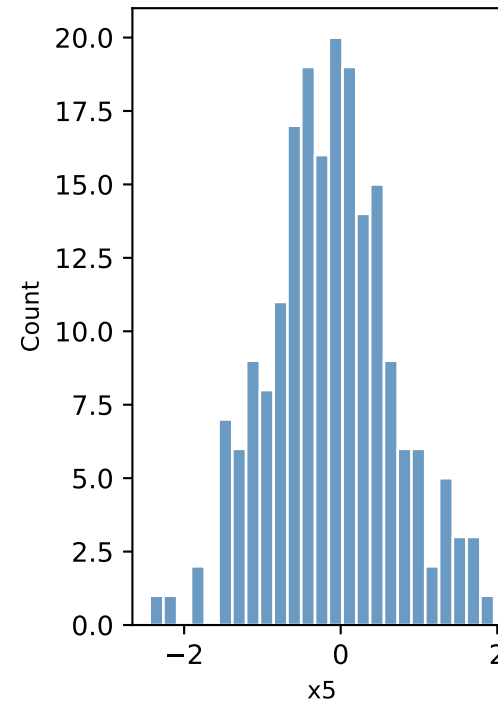
Histogram of x1



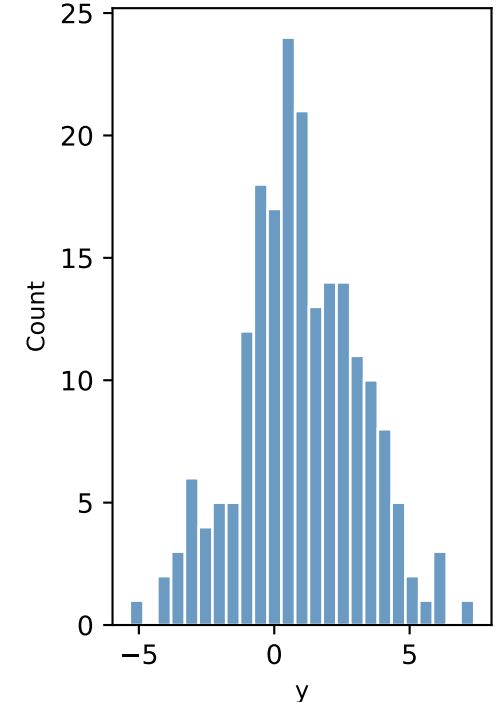
Histogram of x3



Histogram of x5



Histogram of y



Model, Method and Results

Model & Assumptions

Model:
 $y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \epsilon$, $\epsilon \sim N(0, \sigma^2)$

Assumptions:

1. Linearity: y is a linear function of x
2. Independence: observations are i.i.d.
3. Normality: residuals $\sim$ Normal
4. Homoscedasticity: constant variance (equal variance of errors)

Method

Method:
 Ordinary Least Squares (OLS)
 via normal equations

Library / Package:
 numpy – OLS fitting (lstsq)
 scipy.stats – t-test, F-test
 pandas – data handling

Variable Selection Strategy:
 Backward elimination using
 individual p-values ($\alpha=0.05$)

Variable Selection

Variable Selection
 (backward elimination, $\alpha=0.05$)

Full model p-values:

intercept:	p=0.0000	***
x1:	p=0.0000	***
x2:	p=0.0905	<-- drop
x3:	p=0.0000	***
x4:	p=0.7448	<-- drop
x5:	p=0.0000	***

Selected: x1, x3, x5
 Dropped: x2, x4

Reduced Model Formula

Reduced Model (after selection):

$$y = \beta_0 + \beta_1 x_1 + \beta_3 x_3 + \beta_5 x_5 + \epsilon$$

Estimated coefficients:

intercept =	1.5459	(SE=0.0765)
x1 =	0.8702	(SE=0.0626)
x3 =	-1.2954	(SE=0.0553)
x5 =	0.7347	(SE=0.0864)

$\sigma_{\hat{}} = 0.9567$

Coefficients (t-test & p-values)

Name	Coef	SE	t	p-value
intercept	1.5459	0.0765	20.22	0.0000 ***
x1	0.8702	0.0626	13.90	0.0000 ***
x3	-1.2954	0.0553	-23.42	0.0000 ***
x5	0.7347	0.0864	8.50	0.0000 ***

ANOVA, NLL & Summary

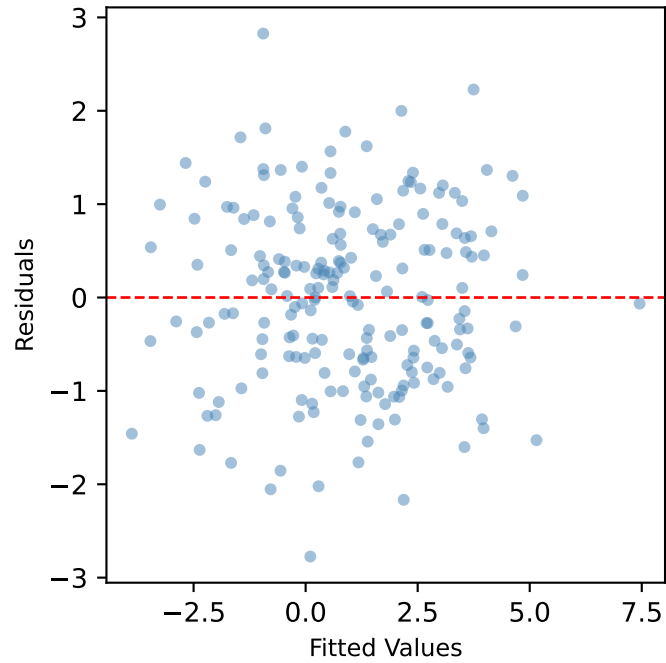
ANOVA (reduced model)
 F = 280.33, p = 1.11e-16
 R-squared: 0.8110
 Adj R-sq: 0.8081

Negative Log-Likelihood (NLL):
 $NLL_i = 0.5 \cdot \log(2 \cdot \pi \cdot \sigma^2) + (y_i - \mu_i)^2 / (2 \cdot \sigma^2)$
 Mean NLL (train) = 1.3646

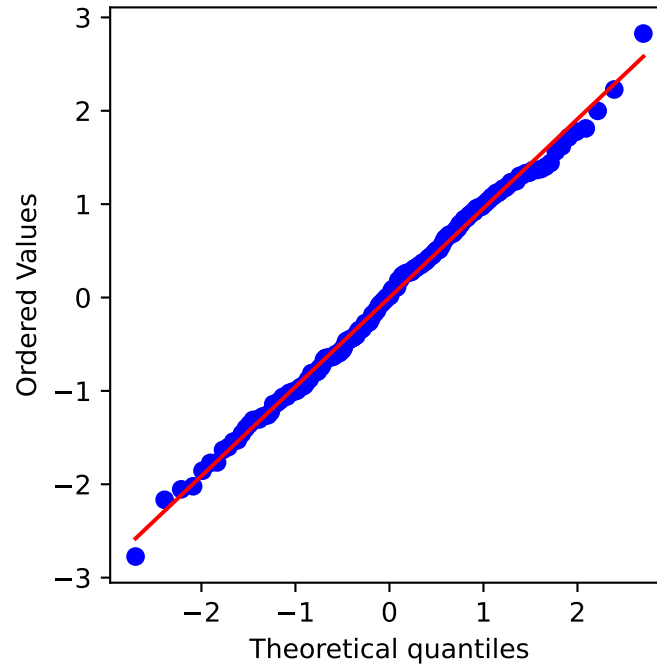
$\sigma_{\hat{}} = 0.9567$
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Visualization

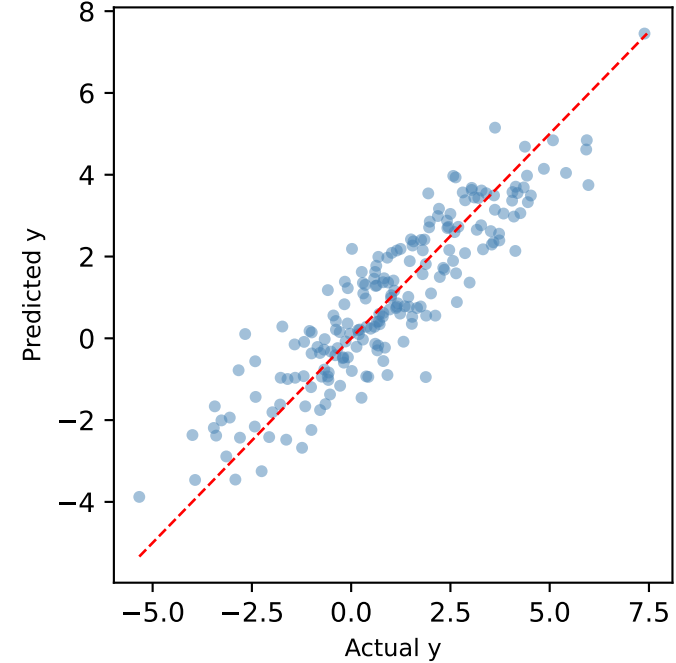
Residuals vs Fitted



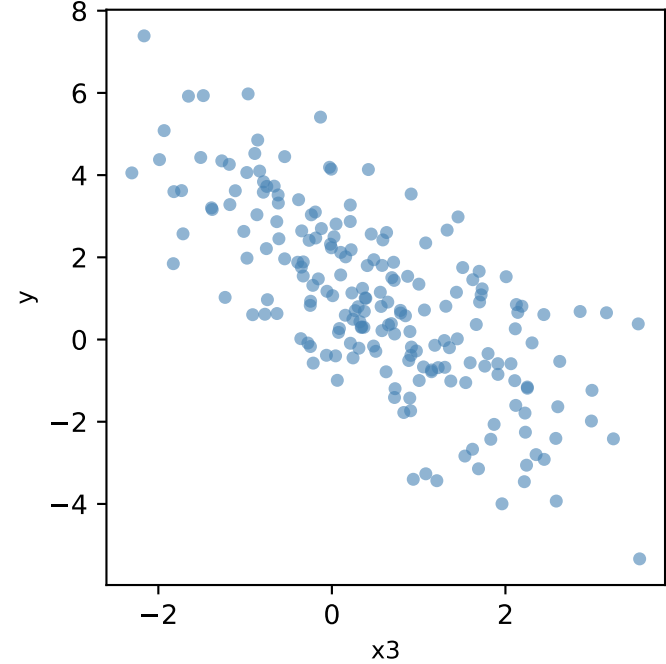
QQ Plot of Residuals



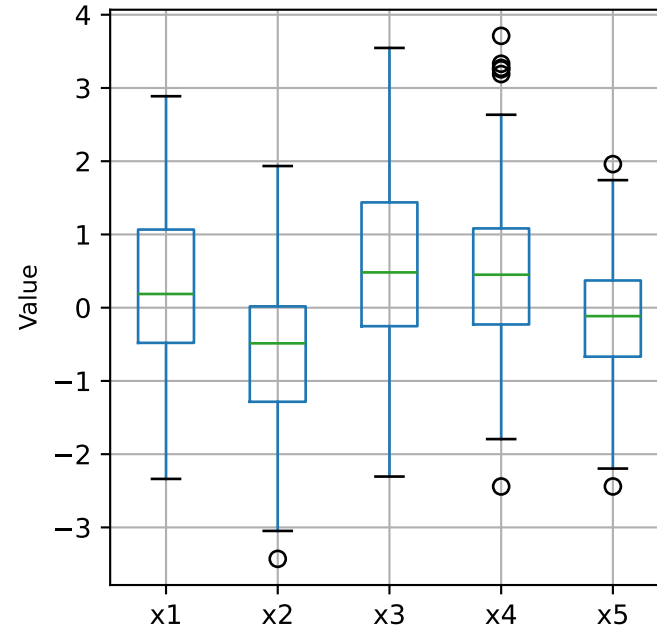
Predicted vs Actual



Scatter: y vs x3



Box Plots of Predictors



Histogram of Residuals

